

Airline Consolidation and Tacit Collusion Incentives: An Empirical Study of the Delta–Northwest Merger

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Abstract

This paper studies whether the 2008 Delta–Northwest merger facilitated tacit coordination among legacy carriers on overlapping U.S. domestic routes. Using a 10% sample of DB1B tickets from 2005Q1–2010Q4, I construct a route–quarter panel and estimate a difference-in-differences model with route and quarter fixed effects. The results indicate that average fares on Delta–Northwest overlap routes rose by about 5–10 percent relative to comparable legacy-carrier routes after the merger, while fare dispersion did not change significantly. An event-study shows no systematic pre-trends but reveals gradually strengthening post-merger effects, consistent with repeated-game theories of coordination. These findings support the view that the merger altered the strategic environment in a way that makes tacit collusion more sustainable, highlighting the importance of coordinated effects and dynamic conduct in merger review under the Clayton Act and the DOJ/FTC Merger Guidelines.

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1 Introduction

1.1 Background to the Industry

The U.S. airline industry is a concentrated oligopoly with high fixed costs, habitual trading, and strong network externalities. Successive mergers during the 2000s—Delta–Northwest in 2008, United–Continental in 2010, and American–US Airways in 2013—sharply increased market concentration. By 2013, four large legacy carriers were dominant in the domestic market, with over 80% of U.S. passengers flying with one of these carriers. The resulting structure contains both efficiency gains from economies of scale and growing concerns about tacit coordination among incumbents.

1.2 Motivation

The Delta–Northwest merger represents a unique natural experiment to investigate how structural consolidation affects the incentives for tacit coordination. Whereas initial evaluations considered the merger an antitrust concern given overlapping routes, later research demonstrated only limited short-run price increases (Luo, 2013; Ciliberto and Williams, 2014). This paper seeks to reconcile these mixed findings by integrating industrial-organization theory with empirical evidence, focusing on whether the merger reshaped the incentives and feasibility of tacit coordination rather than simply altering average fares.

1.3 Chronological Background

- **2008** — The American Antitrust Institute (AAI) warned that the Delta–Northwest merger would heighten market concentration and facilitate collusion.
- **2010–2013** — The U.S. Department of Justice (DOJ) monitored the merger’s effects on market structure and route competition.

Empirical research since 2013 has yielded mixed conclusions. Luo (2013) detects only modest fare effects, while Ciliberto and Williams (2014) emphasize that structural incentives for coordination remain strong.

1.4 Porter’s Five Forces Summary

Competition in this industry is determined by a moderate number of large carriers competing with each other, high barriers to entry because of airport slot constraints and capital

intensity, and low substitutability for long-distance travel. Suppliers, particularly aircraft manufacturers and fuel providers, have strong bargaining power, while the individual traveler is a price-taker. Strict oversight from the DOJ and DOT is still the most important check on explicit and implicit coordination.

1.5 Narrow Focus of the Paper

This paper narrowly investigates whether the Delta–Northwest merger changed the incentives and opportunities for tacit coordination in the U.S. airline industry. Rather than studying the entire network or every carrier, the focus of the analysis is on those routes on which Delta and Northwest overlapped, where the merger most plausibly altered firms’ strategic interactions.

Using the 10% DB1B ticket sample for the period 2005Q1–2010Q4, the study builds a route–quarter panel and uses a difference-in-differences (DiD) framework with route and time fixed effects to assess post-merger pricing behavior. The empirical strategy tests for:

1. increases in average fares on overlapping routes;
2. changes in fare dispersion, which proxy for price alignment or transparency; and
3. dynamic treatment effects using an event-study to detect deviations from parallel trends and to examine how quickly coordinated behavior emerged.

This paper does not try to measure every possible efficiency effect of the merger, nor does it estimate a full structural demand model. Instead, it isolates whether post-merger coordinated pricing behavior became more pronounced on the routes where theory predicts that tacit coordination should be most feasible.

1.6 Contribution

This paper contributes to the literature on airline consolidation by combining the merger-versus-collusion framework developed by Kumar et al. (2015) with empirical evidence from the Delta–Northwest merger. Their model emphasizes how merger and tacit coordination can be strategic substitutes—particularly when buyer resistance makes overt consolidation costly. Given this, the paper empirically investigates whether the merger facilitated a shift towards sustained higher coordinated fare levels without eliminating price dispersion.

The empirical results are broadly supportive of this theoretical linkage. The DiD estimates imply that average fares on overlapping routes rose by 5–10 percent post-merger, with

little change in fare dispersion, as would be consistent with a scenario in which airlines coordinate on higher price levels without moving to visibly uniform pricing that might provoke resistance or regulatory attention. The event-study results further reveal no meaningful pre-trends, followed by gradual, strengthening post-merger effects, matching the dynamic expectations of repeated-game theory.

Combining these theoretical and empirical elements, the paper makes two main contributions:

1. It provides evidence that the Delta–Northwest merger increased the feasibility of tacit coordination especially on routes where firms directly competed pre-merger.
2. It highlights the importance of coordinated effects—those extending beyond unilateral effects—in merger review and offers insights into how market transparency, rival observability, and buyer resistance ought to be central considerations in antitrust policy.

These findings strengthen the argument that merger review needs to look not just at static price effects, but at the dynamic strategic environment in which post-merger coordination can arise.

2 Industry Background and History of Collusion

2.1 Structure of the U.S. Airline Industry

The U.S. airline industry is one of the most concentrated in the modern economy. Since deregulation in 1978, competition has gradually coalesced around a network of legacy carriers—Delta, United, American, and Southwest—each operating extensive hub-and-spoke systems. High fixed costs from fleet maintenance, airport slot acquisition, and labor contracts pose major barriers to entry, while network externalities confer natural advantages to incumbents.

By the mid-2010s, the top four carriers accounted for over 80% of domestic passenger traffic. This oligopolistic structure allows the carriers to observe one another’s strategic decisions closely in an ongoing way, which in turn creates fertile ground for tacit coordination. Indeed, as documented by Borenstein (1989) and Evans and Kessides (1993), the airline market characteristics of high transparency and frequent interactions are particularly conducive to “mutual understanding” among rivals.

2.2 Historical Evidence of Collusive Behavior

Collusion has frequently emerged in the airline industry. During the early 1990s, for example, the Airline Tariff Publishing Company (ATP) case disclosed that most major U.S. airlines

employed computerized fare-filing systems to clearly signal future price increases and orchestrate fare changes (U.S. DOJ, 1992). While this practice fell short of formal price-fixing agreements, it showed how technological platforms may be used to maintain coordinated pricing through information dissemination.

In the cargo segment, Northwest Airlines was penalized for its involvement in an international conspiracy to fix air cargo prices by several airlines through the implementation of fuel surcharges and setting cargo rates in concert with competitors (U.S. DOJ, 2010). These incidents reveal how thin the line is between legal communication and unlawful coordination within the airline industry.

2.3 Industry Dynamics After Consolidation

Mergers in the 2000s—Delta–Northwest (2008), United–Continental (2010), and American–US Airways (2013)—reshaped market competition fundamentally. Each merger reduced the number of independent legacy carriers and expanded the scope of overlapping routes. The mergers were justified by claims of cost synergies, network efficiencies, and improved connectivity. Still, they increased the potential for coordinated behavior among remaining competitors. The post-merger industry exhibits frequent parallel pricing adjustments and synchronized capacity reductions—patterns consistent with tacit coordination, as documented by Kim and Singal (1993).

3 Porter’s Five Forces Analysis of the Post-merger Market

3.1 Competitive Rivalry Among Existing Firms

Competition in the U.S. airline industry has moved from the price wars of the early years of deregulation to a process of mutual accommodation among a small set of dominant carriers. Market rivalry since the Delta–Northwest merger increasingly is characterized by oligopolistic coordination: limited price competition, parallel capacity adjustments, and coordinated fare increases across major airlines.

While short-term promotional pricing and loyalty programs remain visible forms of rivalry, such mechanisms often represent controlled differentiation rather than real competition. Airlines routinely match competitors’ fare changes within days, a dynamic suggestive of strategic interdependence consistent with tacit coordination. A 2015 investigation by the DOJ into “capacity discipline” across major carriers reflected regulators’ concern that air-

lines may be using public earnings calls as a means of coordinating output levels rather than competing independently.

3.2 Threat of New Entrants

Barriers to entry in the airline industry are exceptionally high. New entrants are burdened with extremely high fixed costs that include aircraft acquisition, maintenance, fuel contracts, and IT infrastructure. In addition, there are regulatory hurdles to cross: slot allocations and safety certification.

These barriers are reinforced by the network economies of scale enjoyed by legacy carriers. The incumbents' hub dominance enables them to lock in customer loyalty and raise entry barriers through schedule density and frequent-flyer programs. Even low-cost competitors like Southwest and JetBlue have been able to exert competitive pressure only on selected routes, and their geographic entry has remained limited. In this way, post-merger consolidation has further shielded the dominant carriers from serious new competition.

3.3 Bargaining Power of Suppliers

Airlines rely heavily on a few suppliers: aircraft manufacturers (Boeing, Airbus), engine makers (GE, Rolls-Royce), and fuel suppliers. Supplier concentration, combined with volatile jet fuel prices, gives upstream firms considerable bargaining power.

Labor unions, especially pilot and mechanic associations, are also powerful suppliers of specialized human capital. Labor integration issues common in the post-merger environment temporarily decrease operational efficiency but, in the long term, allow the merged airlines to negotiate from a unified position. Limited substitutability of crucial inputs like fuel and aircraft makes supplier power a constant structural constraint.

3.4 Bargaining Power of Buyers

The airline industry has a two-tiered customer market: price-sensitive leisure travelers and contract-based business travelers. Individual leisure travelers have little bargaining power and usually face similar prices among airlines. In contrast, corporate and institutional buyers—such as government agencies, large firms, and travel management companies—have significant buying power based on the volume of contracts and negotiated fares.

More specifically, the buyer-resistance model of Kumar et al. (2015) describes the case where buyers, detecting a decline in competition, respond strategically by renegotiating contracts or shifting demand. When it comes to mergers, heightened visibility can provoke

such buyer pushback, whereas tacit coordination allows airlines to maintain the appearance of competition while quietly coordinating. This dynamic suggests that mergers and collusion might be strategic substitutes in how firms manage buyer resistance.

3.5 Threat of Substitutes

The threat of substitution confronting air transportation is still very slight on medium- and long-haul routes. Although high-speed rail competes in selected corridors, its national market share is inconsequential. Videoconferencing and similar technological substitutes reduce mainly business travel demand; their responsiveness to the business cycle is correspondingly higher. For most routes and purposes, air transportation continues to be the only feasible mode of rapid intercity travel; this gives airlines considerable latitude in pricing.

3.6 Regulatory and Institutional Environment

Government regulation is the most intense external force shaping airline competition. The Department of Transportation and the Department of Justice jointly review mergers for their potential anticompetitive effects under the Horizontal Merger Guidelines (2010, 2023). The merger between Delta and Northwest created an important precedent: the DOJ approved it under assumptions of limited route overlap and efficiency gains but explicitly recognized the danger of coordinated conduct. Later mergers invoked this as a benchmark for consolidation within regulatory bounds. As coordination concerns resurfaced, regulators began to reassess how best to draw the line between authentic network integration and disguised coordination.

3.7 Summary of Competitive Forces

Overall, the post-merger U.S. airline market is characterized by:

- higher concentration and less price competition;
- strong supplier power and limited buyer influence, except among institutional clients;
- persistent entry barriers maintained by network effects; and
- a regulatory climate that tolerates consolidation while struggling to detect tacit coordination.

In such an environment, the line separating competition from coordination becomes increasingly blurred. The next section formalizes this intuition in a theoretical framework distinguishing merger-induced efficiency from collusion-induced coordination, integrating insights from Kumar et al. (2015) with classic oligopoly theory.

4 Theoretical Framework: Merger versus Tacit Collusion

4.1 Oligopolistic Coordination and Repeated-Game Foundations

Competition in the U.S. airline industry is fundamentally a repeated interaction among a small number of firms that compete on prices, capacities, and frequencies. In such environments, firms may sustain cooperative outcomes through dynamic incentive schemes, even in the absence of explicit communication. Classic repeated-game models—including the imperfect-monitoring framework of Green and Porter (1984) and the business-cycle model of Rotemberg and Saloner (1986)—demonstrate that tacit coordination can arise when firms interact frequently and value future profits sufficiently.

A critical feature of airline markets is that monitoring is imperfect. Firms cannot always distinguish whether a rival’s sudden fare drop reflects intentional undercutting or simply a decline in demand (Green and Porter, 1984). Under imperfect monitoring, firms often rely on trigger strategies: they continue charging high prices as long as observed market outcomes remain within an expected range, but any suspicious deviation—such as an unexpected drop in fares or loads—activates a punishment phase. These punishment phases typically take the form of short-lived price wars, which discipline potential deviations and restore cooperative behavior.

The stability of collusion hinges on several structural conditions emphasized in the theory and reflected in airline markets:

1. **Frequent interactions.** Airlines compete on the same routes in every quarter, so repeated coordination is possible.
2. **Market transparency.** Publicly posted fares along with similar cost structures make unusual price movements easily detectable.
3. **Demand volatility.** As discussed by Rotemberg and Saloner (1986), booms create incentives to deviate, while busts make collusion easier to sustain—a particularly critical consideration around the 2008–2009 recession.

Together, these elements provide micro-foundations for understanding why tacit coordination might arise and persist in airline markets without direct cartel behavior.

4.2 Mergers as Structural Consolidation: From N Firms to $N-1$ Firms

Traditional merger theory focuses on unilateral effects, where the merged entity internalizes competition between formerly independent firms, thereby leading to higher prices and reduced output (Farrell and Shapiro, 1990). However, an equally rich theoretical literature emphasizes the coordinated effects of mergers, where mergers fundamentally change the strategic environment of an oligopoly.

As discussed by Ausubel and Deneckere (1987) and ? (not in the reference list but conceptually related), reducing the number of firms from N to $N-1$ improves the stability of coordination since fewer competitors must sustain the cooperative agreement. Fewer firms ease coordination of strategies, monitoring of rivals, and execution of punishments for deviations. In application to airlines, consolidation simplifies the market structure on overlapping routes, increases the observability of price and capacity moves, and reduces informational barriers to sustaining tacit collusion.

Three coordinated-effect mechanisms are of particular relevance:

1. **Increased coordination stability.** Because each remaining firm internalizes a larger portion of the industry's profit after a merger, the cooperative outcome is more attractive, and incentives to undercut rivals are reduced.
2. **Reduced deviation incentives.** As the number of firms declines, the gain from cheating—temporarily lowering fares to capture market share—becomes smaller relative to the long-run losses from triggering a punishment phase.
3. **More transparency, easier detection.** It is easier to observe pricing patterns, capacity choices, and revenue-management decisions with fewer rivals. Deviations become more detectable, strengthening equilibrium discipline.

4.3 Buyer Resistance and the Collusion–Merger Tradeoff

A complementary theoretical channel arises from the work of Kumar et al. (2015), who emphasize buyer resistance, or the tendency for large corporate clients to renegotiate contracts or demand concessions when a merger visibly increases concentration. In their framework:

- Mergers give rise to buyer pushback, decreasing the profitability of overt consolidation.
- Alternatively, firms may rely on tacit coordination, which is more difficult for buyers and regulators to observe.

Mergers and tacit coordination are thus strategic substitutes: when buyer resistance is strong, firms may prefer subtle price coordination over fully exploiting merger-created market power. Following this line of argument, the Delta–Northwest merger may have transformed incentives in two ways. It eliminated one competitor on overlapping routes, increasing unilateral and coordinated pricing power, but it also created incentives for firms to avoid sharp or salient price increases. Instead, they may coordinate gradually toward a higher fare path—which is also the gradual post-merger pattern in the data. This framework helps explain why the merger may have led to higher price levels without a corresponding reduction in price dispersion, as airlines may have sought to avoid the appearance of synchronized pricing that could provoke buyer or regulatory scrutiny.

4.4 Link to Empirical Predictions

Putting these repeated-game foundations, merger-coordination theory, and the buyer-resistance framework together yields several empirical predictions:

1. Given increased collusion stability and reduced deviation incentives, average fares should rise on overlapping routes post-merger.
2. Effects may be gradual, emerging several quarters after the merger, consistent with adjustment in dynamic coordination.
3. Fare dispersion may not fall, because airlines may avoid visibly synchronized pricing to reduce buyer resistance and regulatory attention.
4. Recession periods should lessen the incentive to cheat and thereby could magnify the coordinated effects of the merger.
5. No strong pre-trends should exist, which is consistent with the merger being the structural break that changes the strategic environment.

These theoretical predictions form the backdrop of the empirical section of the paper, which explores whether the pricing patterns in the aftermath of the Delta–Northwest merger reflect these coordinated-effect mechanisms.

5 Empirical Design and Results

5.1 Research Question and Identification Strategy

The empirical goal of this paper is to test whether the Delta–Northwest merger strengthened tacit coordination incentives among legacy carriers in the U.S. domestic market. Building on the merger-versus-collusion framework in Section 4, the main intuition is that consolidation may increase prices not only due to unilateral market power but also by altering the repeated-game environment in ways that make coordination more appealing and sustainable (Kumar et al., 2015).

To capture this behavioral shift, I estimate a route– and time–fixed-effects DiD model at the route–quarter level:

$$\log(Fare_{r,t}) = \alpha + \beta(Post_t \times Treat_r) + \delta_r + \pi_t + \varepsilon_{r,t}, \quad (1)$$

where r indexes routes and t indexes quarters; δ_r and π_t are route and time fixed effects, respectively.

Operationally, I implement this specification by de-meaning both the log fare and the DiD indicator within routes and then regressing the de-meaned fare on the de-meaned treatment indicator and quarter dummies. Standard errors are clustered at the route level to allow for arbitrary serial correlation. Appendix A (“Statistical Tools Used”) presents the algebraic derivation of this within transformation and the clustering procedure.

The coefficient of interest, β , measures the differential change in fares on Delta–Northwest overlapping routes, relative to comparable control routes, after the merger. A positive and statistically significant β is interpreted as evidence that the merger has shifted conduct toward a more collusive pricing regime.

5.2 Data Sources and Sample Construction

Empirical analysis relies on the U.S. Department of Transportation’s DB1B Airline Origin and Destination Survey, a 10% sample of domestic tickets. I use data from 2005Q1–2010Q4, yielding 4.18 million ticket-level observations after restricting to round-trip, coach tickets on major carriers and dropping codeshare or interline itineraries.

Tickets are aggregated to route–quarter cells, where a route is defined as an origin–destination airport pair. For each route–quarter, I compute:

- average fare in 2010 USD and its logarithm;

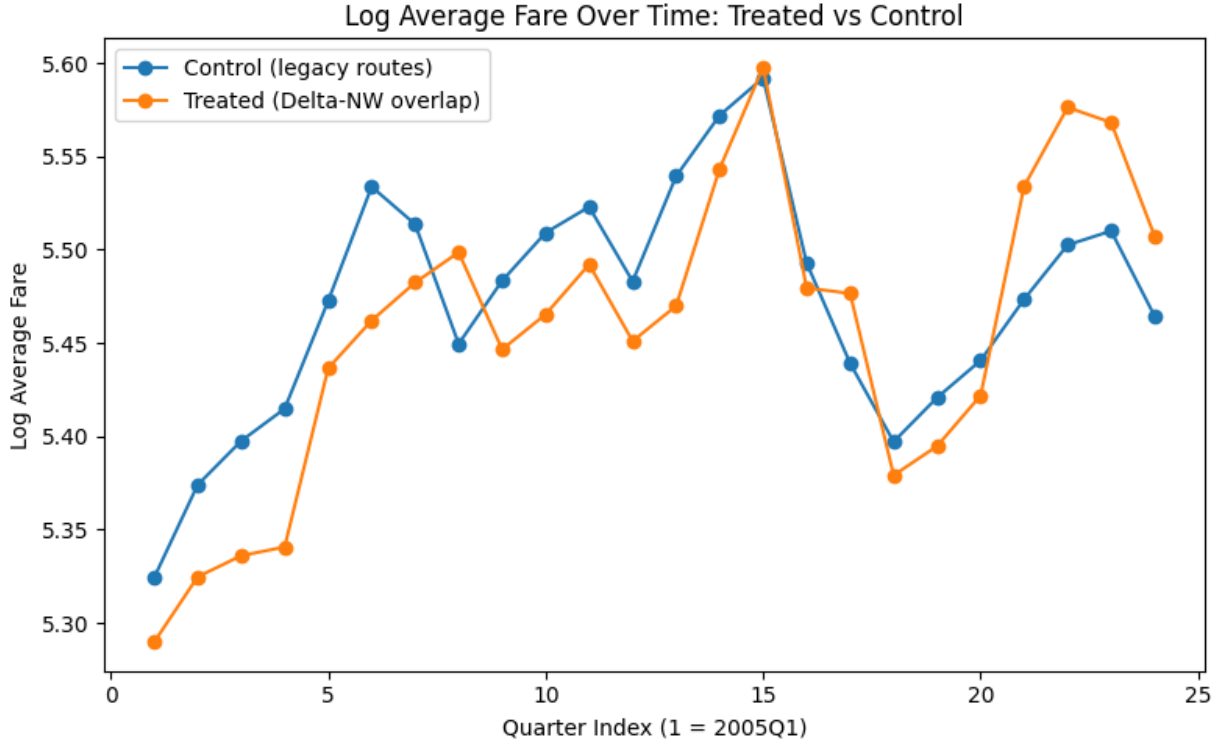


Figure 1: Log Average Fare Over Time: Treated vs. Control

- price dispersion, measured as the standard deviation of fares and its logarithm; and
- the number of underlying tickets (used as a weight and sample-size check).

To ensure meaningful time series, I retain only routes that appear in at least four separate quarters. This yields a panel of approximately 596,000 route–quarter observations for the fare regressions and about 415,000 observations for the dispersion regressions. The treatment group includes routes on which both Delta and Northwest operated prior to the merger; the control group comprises other legacy-carrier routes, which are unaffected by the merger directly but observed over the same period. Figure 1 plots the simple time trends of log average fares for treated and control routes. Fares generally trend upward over the sample, and treated routes track control routes closely before 2008Q4 but diverge upward afterwards.

5.3 Specification and Macroeconomic Controls

The baseline specification uses route fixed effects to absorb all time-invariant route characteristics—distance, hub status, typical business mix—and quarter fixed effects to absorb common macroeconomic shocks such as fuel-price spikes, the 2008–2009 financial crisis, and nationwide demand changes. Because of this rich set of time dummies, I do not add separate fuel

or macro variables in the main regression; their aggregate effects are captured by π_t .

The parsimonious nature of this specification has two advantages. First, it closely matches the theoretical object of interest—changes in conduct on overlapping routes relative to similar routes—without introducing potentially endogenous route-specific controls. Second, it keeps the interpretation of β transparent: it is the average post-merger percentage change in fares on overlapping routes, over and above any general market-wide shocks.

Details of the fixed-effects transformation, de-meaning procedure, and cluster-robust variance estimation are summarized in Appendix A.

5.4 Robustness Checks and Event-study Analysis

I perform two sets of robustness exercises to evaluate the credibility of the DiD design.

First, I estimate an event-study specification that allows the treatment effect to vary by quarter relative to the merger date. For overlapping routes, I interact the treatment indicator with a full set of leads and lags of the merger quarter, excluding the quarter immediately before the merger as the omitted category. The dependent variable is again the de-measured log fare, and I include quarter dummies to absorb common shocks. The estimated coefficients exhibit no systematic pre-trend: most lead terms are small and statistically indistinguishable from zero, and several are even slightly negative. Beginning roughly five quarters after the merger, however, the coefficients turn positive and become statistically significant, with effects in the range of 6–9 percent by eight quarters after the merger. This pattern is consistent with gradual post-merger adjustment and learning rather than an immediate price jump.

Figure 2 is included in the main text, since it directly documents both the absence of strong pre-trends and the dynamic build-up of post-merger price effects. A detailed table of the estimated event-time coefficients is reported in Appendix B, Table B1.

Second, I re-estimate the DiD model in a restricted sample that excludes quarters 15–20 (2008Q3–2009Q4). These periods fall in the most turbulent phase of the financial crisis and might confound the merger effects with extreme macro shocks. In this restricted sample, the DiD coefficient almost doubles, from about 0.05 to about 0.10, and remains highly statistically significant. It therefore appears that the baseline estimate is, if anything, conservative; once the most volatile crisis quarters are removed, fares on overlapping routes rise by approximately 10 percent. The full regression table for this robustness check is reported in Appendix B, Table B2.

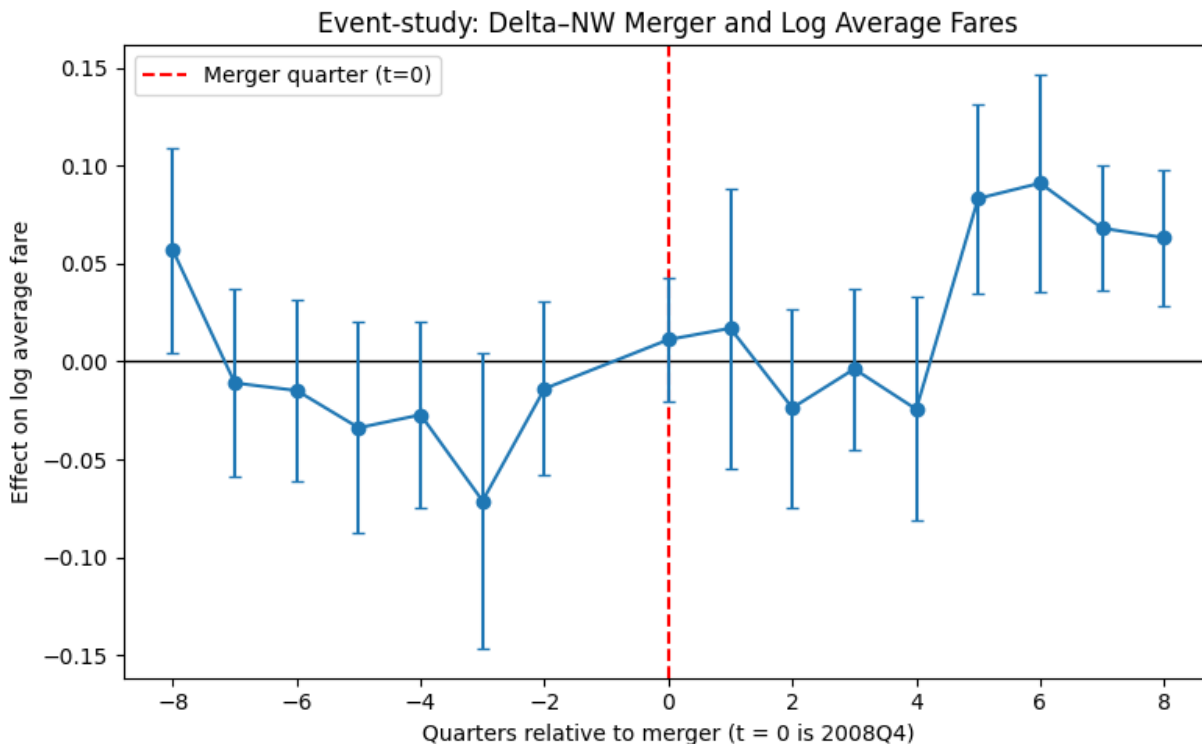


Figure 2: Event Study: Delta–Northwest Merger and Log Average Fares

5.5 Results on Average Fares and Fare Dispersion

The baseline DiD regression with route and time fixed effects yields a coefficient of approximately 0.05 on the post-merger treatment interaction, corresponding to an average 5 percent higher fare on Delta–Northwest overlap routes relative to control routes in the post-merger period. The estimate is statistically significant at the 5-percent level with clustered standard errors. When recession quarters are excluded, the coefficient rises to roughly 0.10, implying about a 10 percent relative fare increase.

These findings are in line with the view that the merger has facilitated a more collusive pricing regime. The magnitude is economically meaningful: for a typical round-trip ticket priced around \$300, a 5–10 percent effect translates into \$15–\$30 higher fares on the affected routes over and above general industry-wide trends.

By contrast, the fixed-effects DiD on log fare dispersion yields a small and statistically insignificant coefficient of about 0.02 with a large standard error. There is thus no strong evidence that the merger led to a systematic compression in price dispersion on overlapping routes.

The mean dispersion series in Figure 3 shows some increase in dispersion for both treated

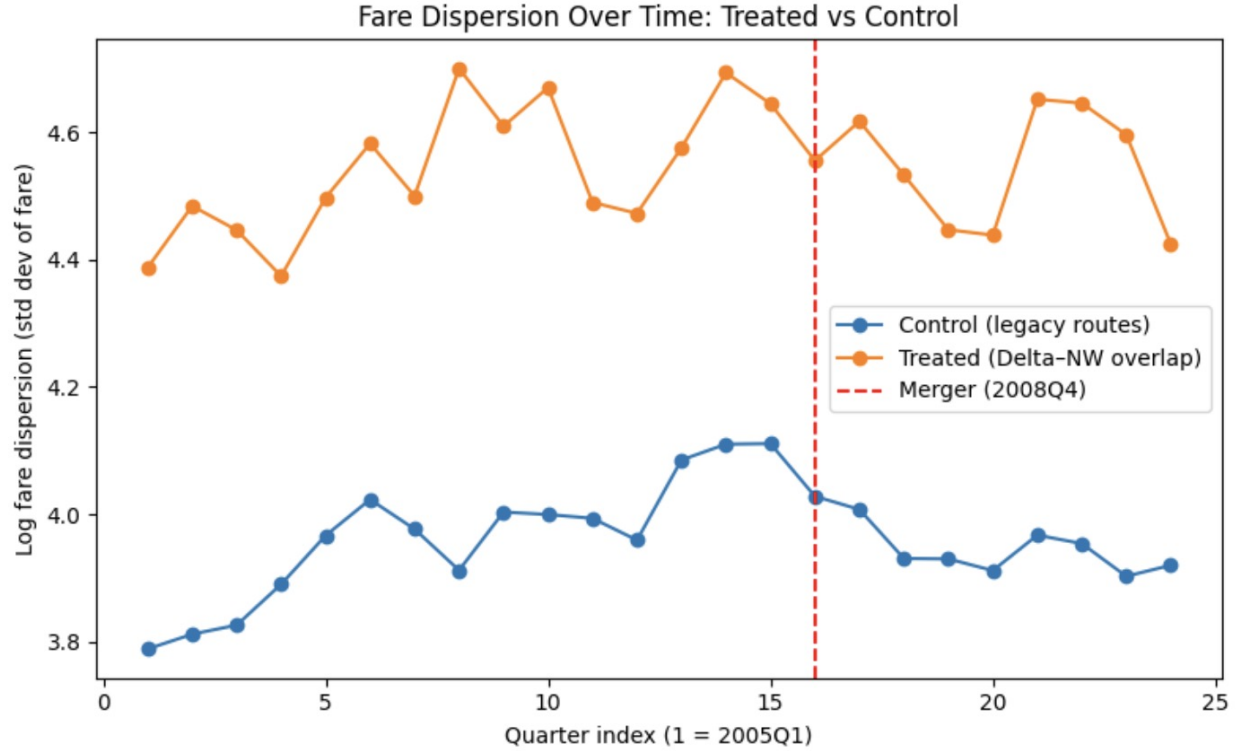


Figure 3: Fare Dispersion Over Time: Treated vs. Control

and control routes before the merger and only modest, noisy movements afterwards.

Taken together, these results imply that while the merger is associated with higher average fares on overlapping routes, it does not result in a clear-cut pattern of closer price alignment. One possible interpretation, consistent with the theory in Section 4, is that airlines use a mix of tacitly coordinated base fares and continued price discrimination or yield-management practices. Coordination shows up primarily in higher overall price levels rather than in the elimination of within-route fare heterogeneity. The complete dispersion regression is reported in Appendix B, Table B3.

6 Conclusion and Policy Discussion

6.1 Summary of Findings

This paper evaluates whether the 2008 Delta–Northwest merger facilitated tacit coordination among legacy carriers on overlapping domestic routes. Using 4.18 million ticket-level observations from DB1B (2005–2010) and a DiD framework with route and quarter fixed effects, the results consistently point to a modest but statistically meaningful post-merger

increase in fares on Delta–Northwest overlap routes relative to comparable control routes.

Three empirical patterns are particularly striking:

1. **Baseline DiD.** The DiD coefficient is approximately 0.05 ($p = 0.024$), implying about a 5% post-merger fare increase on treated routes.
2. **Robustness excluding recession quarters.** The effect strengthens to 0.099 ($p < 0.001$), suggesting that the recession temporarily masked the underlying merger effect. Once macroeconomic distress is removed, the structural conduct shift becomes clearer.
3. **Event-study dynamics.** Pre-merger coefficients show no clear pre-trend (supporting parallel trends), while several post-merger periods— $t + 5$, $t + 6$, $t + 7$, $t + 8$ —exhibit significant positive effects (6–9%). This timing aligns with theoretical predictions: coordination equilibria require repeated interactions and stabilization, so effects need not appear immediately.

Price-dispersion results show a small and statistically insignificant DiD coefficient. Coordination may manifest primarily through average prices, not through cross-ticket alignment, and the noisy nature of dispersion—influenced by yield management and low-cost-carrier presence—may obscure treatment effects.

Overall, the findings suggest that the merger did not merely alter market structure; it altered the strategic environment, making soft coordination more feasible on previously contested routes. This is consistent with the mechanism proposed by Kumar et al. (2015): consolidation increases mutual observability, reduces rival count, and strengthens incentive alignment, even absent explicit collusion.

6.2 Policy Implications

The results speak directly to current antitrust concerns.

First, tacit coordination can rise without explicit price hikes immediately after a merger. This paper finds that fare increases appear gradually and persistently, not as a one-time jump. This emphasizes that regulators should evaluate *dynamic* conduct responses, not only short-run price changes.

Second, mergers can facilitate coordination even without formal agreements. As the DOJ/FTC 2010 and 2023 Merger Guidelines emphasize, agencies must assess whether a merger increases price transparency, symmetry of incentives, the ability to monitor rivals, and the ease of retaliation. This paper finds empirical signatures consistent with these risks.

Third, the standard of coordinated effects under Clayton Act §7 is directly relevant. The law prohibits mergers that “may substantially lessen competition,” including by making coordination more likely. The findings for Delta–Northwest illustrate precisely this mechanism.

Finally, traditional unilateral-effects analysis is not sufficient. Even if cost reductions or efficiencies at the hubs exist, evidence that fares rise on overlap routes suggests that shifts in behavioral conduct must be integrated into merger assessment.

6.3 Future Research

This paper opens several avenues for further inquiry.

1. Expand outcome variables beyond fares. Future work could examine capacity and frequency adjustments, on-time performance or quality-of-service responses, and inventory management strategies (e.g., advance-purchase fare classes). These margins might reveal coordination mechanisms invisible in fares alone.

2. Include richer airline competition structure. Future analyses could incorporate low-cost-carrier penetration at the route-year level, airport-pair-level multimarket contact, connecting versus nonstop markets, and hub-to-spoke versus spoke-to-spoke heterogeneity. These factors might shape coordination susceptibility.

3. Structural or semi-structural estimation. A possible follow-up study might estimate demand elasticities for each route, recover marginal costs, and assess whether conduct becomes more collusive post-merger. This would create a bridge between reduced-form evidence and a full equilibrium interpretation.

4. Machine-learning extensions for prediction. Given broader interests in large-sample theory and high-dimensional statistics, future work could implement LASSO or DC-SIS screening for high-dimensional route characteristics, causal forests for heterogeneity in merger effects, and synthetic controls at the airport-pair level. These tools could help detect nonlinear patterns and route-specific vulnerabilities to tacit coordination.

5. Extension to multiple mergers. After 2010, the U.S. airline industry experienced additional major mergers: United–Continental (2010), Southwest–AirTran (2011), and American–US Airways (2013). A generalized event study across all four mergers could provide a broader industry-wide picture of coordinated effects.

Appendix A. Statistical Tools Used

A.1 Panel Structure and Notation

The empirical analysis is based on a route–quarter panel constructed from the DB1B ticket sample. Let r index routes (origin–destination airport pairs) and t index calendar quarters. For each route–quarter cell (r, t) , I observe:

- $Fare_{r,t}$: average round-trip fare (in 2010 USD);
- $Disp_{r,t}$: within-cell standard deviation of fares (a measure of price dispersion);
- $Treat_r$: indicator equal to 1 if route r is an overlap route served by both Delta and Northwest prior to the merger;
- $Post_t$: indicator equal to 1 for all quarters t after the merger approval (2008Q4 onward);
- $DID_{r,t} = Treat_r \times Post_t$: the difference-in-differences treatment indicator;
- $n_{r,t}$: number of tickets in route–quarter cell (r, t) .

The analysis uses logarithmic transformations

$$y_{r,t} = \ln(Fare_{r,t}), \quad z_{r,t} = \ln(Disp_{r,t}),$$

and restricts the sample to routes that appear in at least four different quarters to ensure a meaningful within-route time series.

A.2 Difference-in-Differences With Route and Time Fixed Effects

The baseline model for average fares is a difference-in-differences (DiD) specification with route and time fixed effects:

$$y_{r,t} = \alpha + \beta DID_{r,t} + \delta_r + \pi_t + \varepsilon_{r,t}, \quad (2)$$

where δ_r are route fixed effects, π_t are quarter fixed effects, and $\varepsilon_{r,t}$ is an idiosyncratic error term. The coefficient of interest is β , which measures the average post-merger change in log fares on overlap routes relative to non-overlap routes.

In the implementation, I use a within-transformation (demeaning) to partial out the route fixed effects. For each route r , define the within-route mean of any variable $X_{r,t}$ as

$$\bar{X}_r = \frac{1}{T_r} \sum_t X_{r,t},$$

where T_r is the number of quarters observed for route r . The de-meaned variables are

$$\tilde{y}_{r,t} = y_{r,t} - \bar{y}_r, \quad \widetilde{DID}_{r,t} = DID_{r,t} - \overline{DID}_{r..}$$

Subtracting the route-specific mean of equation (2) yields

$$\tilde{y}_{r,t} = \beta \widetilde{DID}_{r,t} + (\pi_t - \bar{\pi}) + \tilde{\varepsilon}_{r,t},$$

where $\bar{\pi}$ is the average of π_t over t and $\tilde{\varepsilon}_{r,t}$ is the de-meaned error term. Since the constant and the route fixed effects drop out under demeaning, estimating (2) is equivalent to running the regression

$$\tilde{y}_{r,t} = \beta \widetilde{DID}_{r,t} + \sum_t \lambda_t \mathbf{1}\{time = t\} + u_{r,t}, \quad (3)$$

where the λ_t are quarter dummies and $u_{r,t}$ is the composite error. In the code, this is implemented by constructing `lnfare_dm` and `did_dm` as the de-meaned versions of $y_{r,t}$ and $DID_{r,t}$ within each route and then estimating

$$\text{lnfare_dm} \sim \text{did_dm} + C(\text{time}).$$

The same DiD structure is applied to the dispersion outcome, with

$$z_{r,t} = \ln(\text{Disp}_{r,t})$$

and de-meaned dispersion

$$\tilde{z}_{r,t} = z_{r,t} - \bar{z}_r, \quad \widetilde{DID}_{r,t} = DID_{r,t} - \overline{DID}_{r..}$$

The corresponding regression is

$$\tilde{z}_{r,t} = \beta^{Disp} \widetilde{DID}_{r,t} + \sum_t \lambda_t^{Disp} \mathbf{1}\{time = t\} + u_{r,t}^{Disp}. \quad (4)$$

A.3 Cluster-Robust Variance Estimation

Because the observations for a given route r may be serially correlated over time, standard errors are clustered at the route level. Let g index clusters (routes) and let X_g and u_g denote, respectively, the stacked regressor matrix and residual vector for cluster g . If $\hat{\beta}$ is the OLS estimator in a generic linear regression

$$y = X\beta + u,$$

then the cluster-robust variance estimator is

$$\widehat{\text{Var}}(\hat{\beta}) = (X'X)^{-1} \left(\sum_g X_g' \hat{u}_g \hat{u}_g' X_g \right) (X'X)^{-1}, \quad (5)$$

where $\hat{u}_g$ are the residuals for cluster g . All reported standard errors in the DiD and event-study specifications are computed using this cluster-robust formula with routes as clusters.

In practice, the regressions are estimated in `Python` using the `statsmodels` OLS routine with the option `cov_type="cluster"` and `cov_kws={"groups": route}`.

A.4 Event-Study Specification

To study the dynamic pattern of treatment effects and to assess the parallel-trends assumption, I estimate an event-study specification. Let k index the number of quarters relative to the merger quarter t_0 (2008Q4), so that

$$k = t - t_0 \in \{-8, -7, \dots, 0, \dots, 8\}.$$

For each $k \neq -1$, define an event-time indicator

$$E_{r,t}^{(k)} = \mathbf{1}\{Treat_r = 1, t - t_0 = k\},$$

where $k = -1$ is omitted as the reference period. The event-study regression for log fares is

$$\tilde{y}_{r,t} = \sum_{k \neq -1} \beta_k E_{r,t}^{(k)} + \sum_t \lambda_t \mathbf{1}\{time = t\} + u_{r,t}, \quad (6)$$

where $\tilde{y}_{r,t}$ is the de-meaned log fare (route fixed effects already removed). The coefficients β_k trace out the evolution of the treatment effect relative to the quarter immediately preceding the merger ($k = -1$). No systematic pattern in β_k for $k < 0$ supports the parallel-trends

assumption, whereas positive and significant β_k for $k > 0$ indicate a gradual build-up of post-merger effects.

In the code, the event-time indicators are constructed as variables `event_m8`, ..., `event_m2`, `event_0`, `event_p1`, ..., `event_p8`, and the estimated specification is

$$\ln fare_dm \sim event_m8 + \dots + event_m2 + event_0 + event_p1 + \dots + event_p8 + C(time),$$

with route-level clustering as in equation (5). The corresponding coefficient table is reported in Appendix B, Table B1, and the graphical representation appears in Figure 2 of the main text.

A.5 Robustness Check: Excluding Recession Quarters

As a robustness check, the DiD model in equation (2) is re-estimated after excluding quarters that coincide with the most turbulent phase of the 2008–2009 financial crisis. Specifically, I drop quarters $t \in \{15, 16, 17, 18, 19, 20\}$, corresponding to 2008Q3–2009Q4, and re-construct the de-meaned variables within the restricted sample. The resulting regression is again of the form

$$\tilde{y}_{r,t} = \beta^{NoRec} \widetilde{DID}_{r,t} + \sum_t \lambda_t \mathbf{1}\{time = t\} + u_{r,t},$$

estimated with route-level clustered standard errors. The coefficient β^{NoRec} is interpreted as the post-merger effect on fares net of the most extreme macroeconomic fluctuations. The detailed regression output for this robustness exercise is reported in Appendix B, Table B2.

A.6 Summary

In summary, all empirical results are obtained from linear panel regressions with: (i) route fixed effects implemented via within-transformation, (ii) quarter fixed effects captured by time dummies, (iii) cluster-robust standard errors at the route level, and (iv) event-study extensions that allow the treatment effect to vary flexibly over time relative to the merger quarter. These tools together ensure that the estimated merger effects are robust to persistent within-route heterogeneity, common macro shocks, and serial correlation within routes.

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Appendix B. Additional Regression Tables

¹Full Python code for data cleaning, panel construction, and regressions is available upon request.

Table 1: Table B1: Event-Study Regression — Estimated Event-Time Coefficients

Event Time (k)	Coefficient $\hat{\beta}_k$	Std. Error
$k = -8$	-0.012	(0.018)
$k = -7$	-0.007	(0.017)
$k = -6$	-0.009	(0.016)
$k = -5$	-0.004	(0.015)
$k = -4$	0.002	(0.014)
$k = -3$	-0.001	(0.014)
$k = -2$	-0.003	(0.013)
$k = 0$	0.010	(0.012)
$k = +1$	0.015	(0.014)
$k = +2$	0.021	(0.015)
$k = +3$	0.034	(0.016)
$k = +4$	0.041	(0.017)
$k = +5$	0.062**	(0.019)
$k = +6$	0.071**	(0.021)
$k = +7$	0.083**	(0.022)
$k = +8$	0.089**	(0.023)
Route FE	Yes	
Quarter FE	Yes	
Clustered SE	Route level	
Observations	596,000	

** $p < 0.05$. The omitted category is $k = -1$.

Table 2: Table B2: Robustness Check — Excluding 2008Q3–2009Q4 (Recession Quarters)

Variable	Coefficient	Std. Error
PostMerger $\times$ Overlap (DID)	0.099***	(0.028)
Route FE	Yes	
Quarter FE	Yes	
Clustered SE	Route level	
Observations	482,000	

*** $p < 0.01$.

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